MA1522 Linear Algebra for Computing

Lesson 11: Coordinates and Transition Matrices

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson11` (e.g. `A1234567Z_lesson11.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 11**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Section 3.6 Coordinates Relative to a Basis

Definition

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a basis for V, a subspace of $\mathbb{R}^n$, and let $\mathbf{v} \in V$ have the unique expression $\mathbf{v} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k$. The vector $[\mathbf{v}]_S = (c_1, \ldots, c_k)^T \in \mathbb{R}^k$ is called the **coordinates of v relative to the basis** S.

Algorithm

To find $[\mathbf{v}]_S$, solve the linear system $\left(\begin{array}{cccc|c} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k & \mathbf{v} \end{array}\right)$.

Note that $\mathbf{v} \in \mathbb{R}^n$ but $[\mathbf{v}]_S \in \mathbb{R}^k$, and that $[\mathbf{v}]_S$ depends on the ordering of S.

Example 1

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for a subspace $V \subseteq \mathbb{R}^5$, and let $\mathbf{v} \in V$ be such that

$$\operatorname{rref}\left(\begin{array}{ccc|c} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{v} \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

What is $[\mathbf{v}]_S$?

$[\mathbf{v}]_S = (1, -5, 0)^T \in \mathbb{R}^3$, not $\mathbb{R}^5$: S has three vectors, so $[\mathbf{v}]_S$ has three entries.

Question 1

Let V be the subspace of $\mathbb{R}^4$ with basis $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, where

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

(a) Show that $\mathbf{v} = (-1, 1, 3, 1)^T$ lies in V, and find $[\mathbf{v}]_S$.

(b) Show that $\mathbf{w} = (2, 1, 1, 1)^T$ does not lie in V.

(c) Suppose $\mathbf{x} \in V$ has $[\mathbf{x}]_S = (2, -1, 3)^T$. Find $\mathbf{x}$.

Question 1 — solution

(a) and (b) Put $\mathbf{v}$ and $\mathbf{w}$ after the bar and reduce once.

$$\left(\begin{array}{ccc|cc} 1 & 0 & 0 & -1 & 2 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right) \xrightarrow[R_4 - R_3]{R_2 - R_1,\ R_3 - R_2} \left(\begin{array}{ccc|cc} 1 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 2 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & -1 \end{array}\right).$$

The $\mathbf{v}$ column is consistent, so $\mathbf{v} \in V$ and $[\mathbf{v}]_S = (-1, 2, 1)^T$: indeed $-\mathbf{u}_1 + 2\mathbf{u}_2 + \mathbf{u}_3 = (-1, 1, 3, 1)^T = \mathbf{v}$. ✓

The $\mathbf{w}$ column has a pivot in the last row, so that system is inconsistent: $\mathbf{w} \notin \operatorname{span}(S) = V$.

(c) Read the definition forwards: $\mathbf{x} = 2\mathbf{u}_1 - \mathbf{u}_2 + 3\mathbf{u}_3 = (2, 1, 2, 3)^T$. ✓ □

Section 3.6 Properties of Coordinates

Proposition

Let S be a basis for a subspace V of $\mathbb{R}^n$. Then coordinates **preserve linear combinations**: for all $\mathbf{v}_1, \ldots, \mathbf{v}_m \in V$ and all scalars $c_1, \ldots, c_m$,

$$[\, c_1\mathbf{v}_1 + \cdots + c_m\mathbf{v}_m \,]_S = c_1[\mathbf{v}_1]_S + \cdots + c_m[\mathbf{v}_m]_S.$$

Moreover the correspondence $\mathbf{v} \leftrightarrow [\mathbf{v}]_S$ is one-to-one: if $[\mathbf{x}]_S = [\mathbf{y}]_S$ then $\mathbf{x} = \mathbf{y}$, since the expression of a vector in a basis is unique.

Example 2

True or false? Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a basis for $V \subseteq \mathbb{R}^n$, and let $\mathbf{w}_1, \ldots, \mathbf{w}_l$ and $\mathbf{v}$ be vectors in V.

1. $\{\mathbf{w}_1, \ldots, \mathbf{w}_l\}$ is linearly independent if and only if $\{[\mathbf{w}_1]_S, \ldots, [\mathbf{w}_l]_S\}$ is linearly independent.

2. $\mathbf{v} \in \text{span}\{\mathbf{w}_1, \ldots, \mathbf{w}_l\}$ if and only if $[\mathbf{v}]_S \in \text{span}\{[\mathbf{w}_1]_S, \ldots, [\mathbf{w}_l]_S\}$.

Both are true, and both come from one equivalence: for any $\mathbf{x} \in V$ and any scalars $c_1, \ldots, c_l$,

$$\mathbf{x} = c_1\mathbf{w}_1 + \cdots + c_l\mathbf{w}_l \iff [\mathbf{x}]_S = c_1[\mathbf{w}_1]_S + \cdots + c_l[\mathbf{w}_l]_S.$$

($\Rightarrow$) take coordinates and apply the Proposition. ($\Leftarrow$) the right-hand side is $[\, c_1\mathbf{w}_1 + \cdots + c_l\mathbf{w}_l \,]_S$, and equal coordinates force equal vectors.

(1) Put $\mathbf{x} = \mathbf{0}$: the two homogeneous equations have the same solutions $(c_1, \ldots, c_l)$, so one has only the trivial solution exactly when the other does.

(2) Put $\mathbf{x} = \mathbf{v}$: such scalars exist on one side exactly when they exist on the other. $\square$

Question 2

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be the basis for $V \subseteq \mathbb{R}^4$ of Question 1, and let $T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq V$ be such that

$$\begin{pmatrix} [\mathbf{v}_1]_S & [\mathbf{v}_2]_S & [\mathbf{v}_3]_S \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix}.$$

(a) Show that T is linearly independent.

(b) Express $\mathbf{u}_1$ as a linear combination of the vectors in T.

Question 2 — solution

(a) The three coordinate vectors are the columns of a square matrix with

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix} = 1 \neq 0,$$

so they are linearly independent in $\mathbb{R}^3$. By Example 2, T is linearly independent.

(b) Write $\mathbf{u}_1 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$ and take coordinates relative to S of both sides. Since $[\mathbf{u}_1]_S = \mathbf{e}_1$, the c_i solve

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 0 \\ 1 & 1 & 2 & 0 \end{array}\right) \xrightarrow{R_2-R_1,\ R_3-R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right) \xrightarrow[R_2-2R_3]{R_1-R_2,\ R_1+R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right).$$

So $\mathbf{u}_1 = \mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3$. Check: $[\mathbf{v}_1]_S + [\mathbf{v}_2]_S - [\mathbf{v}_3]_S = (1,0,0)^T = [\mathbf{u}_1]_S$. ✓ □

Section 3.8 Transition Matrices

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $T = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be bases for a subspace V of $\mathbb{R}^n$.

Definition

The **transition matrix from T to S** is $P = \begin{pmatrix} [\mathbf{v}_1]_S & [\mathbf{v}_2]_S & \cdots & [\mathbf{v}_k]_S \end{pmatrix}$.

Theorem

For every $\mathbf{w} \in V$, $[\mathbf{w}]_S = P[\mathbf{w}]_T$.

Algorithm

$\begin{pmatrix} \mathbf{u}_1 \cdots \mathbf{u}_k & | & \mathbf{v}_1 \cdots \mathbf{v}_k \end{pmatrix} \longrightarrow \begin{pmatrix} I_k & P \\ 0 & 0 \end{pmatrix}$: the basis you want to land in goes on the left.

Question 3

As in Question 2, let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for $V \subseteq \mathbb{R}^4$ and let $T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq V$ be such that

$$\begin{pmatrix} [\mathbf{v}_1]_S & [\mathbf{v}_2]_S & [\mathbf{v}_3]_S \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix}.$$

(a) Write down the transition matrix P from T to S.

(b) Find the transition matrix Q from S to T.

(c) Compute PQ, and say what it tells you about P and Q.

Question 3 — solution: (a) and (b)

(a) By definition the transition matrix from T to S is $\big([\mathbf{v}_1]_S \; [\mathbf{v}_2]_S \; [\mathbf{v}_3]_S \big)$, which is the matrix printed in Question 2:

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix}.$$

(b) Now T is the basis to land in, so S goes after the bar:

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right),$$

so Q has columns $[\mathbf{u}_1]_T, [\mathbf{u}_2]_T, [\mathbf{u}_3]_T$. Its first column is $(1, 1, -1)^T$, which is $\mathbf{u}_1 = \mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3$ from Question 2(b). ✓

Question 3 — solution: (c)

Multiplying out,

$$PQ = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -2 \\ -1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

So P is invertible and $Q = P^{-1}$: converting a coordinate vector into S and then back into T changes nothing. □

Challenge

Let S and T be bases for a subspace V of $\mathbb{R}^n$, and let P be the transition matrix from T to S.

Show that P is invertible, and that P^{-1} is the transition matrix from S to T.

You may not assume that P is invertible — that is what is being proved.

Next lesson

- **Sections 4.1 & 4.2 Column Space, Row Space, Nullspace, and Rank**

Please watch the video clips for these sections beforehand.

Challenge — solution

Write $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$. Since T is also a basis for V, the transition matrix Q from S to T exists, and by the same theorem

$$[\mathbf{w}]_T = Q[\mathbf{w}]_S \quad \text{for every } \mathbf{w} \in V.$$

Combining the two, for every $\mathbf{w} \in V$,

$$[\mathbf{w}]_S = P[\mathbf{w}]_T = PQ[\mathbf{w}]_S.$$

Now every $\mathbf{c} = (c_1, \ldots, c_k)^T \in \mathbb{R}^k$ arises this way: put $\mathbf{w} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k$, which lies in V and has $[\mathbf{w}]_S = \mathbf{c}$. Hence $PQ\mathbf{c} = \mathbf{c}$ for every $\mathbf{c} \in \mathbb{R}^k$, that is,

$$PQ = I_k.$$

A square matrix with a right inverse is invertible (Chapter 2), so P is invertible and $P^{-1} = Q$, the transition matrix from S to T. □